Supplementary Materials
Large, Complex, and Realistic Safety Clothing and Helmet Detection: Dataset and Method

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ABSTRACT

Detecting safety clothing and helmet is paramount for ensuring the safety of construction workers. However, the scarcity of high-quality datasets has impeded the development of deep learning models in this domain. In this study, we construct a large, complex, and realistic safety clothing and helmet detection (SFCHD) dataset. SFCHD is derived from two authentic chemical plants, comprising 12,373 images, 7 categories, and 50,552 annotations. We partition the SFCHD dataset into training and testing sets with a ratio of 4:1 and validate its utility by applying several classic object detection algorithms. Furthermore, drawing inspiration from the attention mechanism, we design a spatial and channel attention-based low-light enhancement (SCALE) module. SCALE is a plug-and-play component with a high degree of flexibility. Extensive evaluations of the SCALE module on both the low-light ExDark and contributed SFCHD datasets have empirically demonstrated its efficacy in enhancing the performance of detectors under low-light conditions. The dataset and code are publicly available at https://github.com/lijfrank-open/SFCHD-SCALE.

CCS CONCEPTS

• **Computing methodologies → Object detection**.

KEYWORDS

Computer Industry, Object Detection, Image Processing, Construction Safety

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A MORE IMAGES

A.1 Images in the SFCHD Dataset

The SFCHD dataset encompasses 12,373 images across 40 different scenarios, covering 7 categories with a total of 50,552 annotated instances. These images are sourced from authentic construction sites, ensuring the practical applicability of the data. All images are captured from factory surveillance cameras, encompassing two chemical plants. To our knowledge, SFCHD is the inaugural factory dataset that concurrently satisfies the detection of safety clothing and helmet. Fig. 1 illustrates images from the SFCHD dataset.

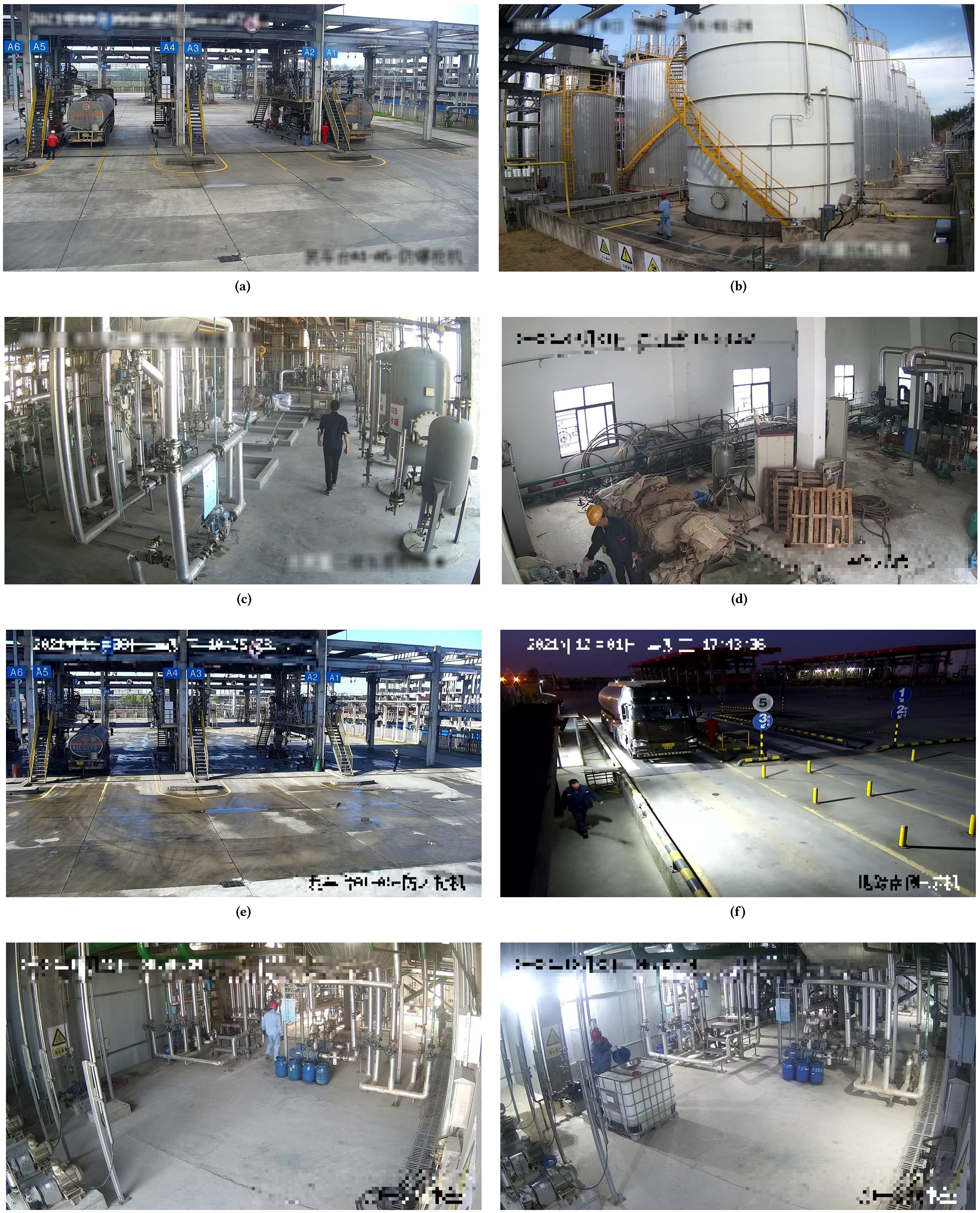
A.2 Images With Different Lighting

As illustrated in Fig. 2, we categorize the entire dataset into three groups: high-light, low-light, and normal-light images, based on the intensity of light sources around person in the images. Here, (a)-(d), (e)-(h), and (i)-(l) are the images based on normal-light, high-light, and low-light, respectively. In (e)-(h), it is extremely difficult to detect whether the worker is wearing safety clothing and helmet because the surrounding light is too high. Similarly, in (i)-(l), too low light around the worker makes it difficult to conduct object detection tasks.

B MORE RELATED WORKS

B.1 Object Detection Dataset

The most remarkable datasets in the field of object detection include Pascal VOC [6, 7], MS-COCO [20], ILSVRC [34], and Open Image [17]. The Pascal Vision Object Class (VOC) Challenge is a competition to accelerate the advancement of the field of computer vision, and two versions of the datasets adopted in the competition, VOC 2007 and VOC 2012, were commonly used for standard benchmarks. The Microsoft Common Objects in Context (MS-COCO) is



(a) (b) (c) (d) (e) (f) (g) (h)

Figure 1: Image examples from the SFCHD dataset.

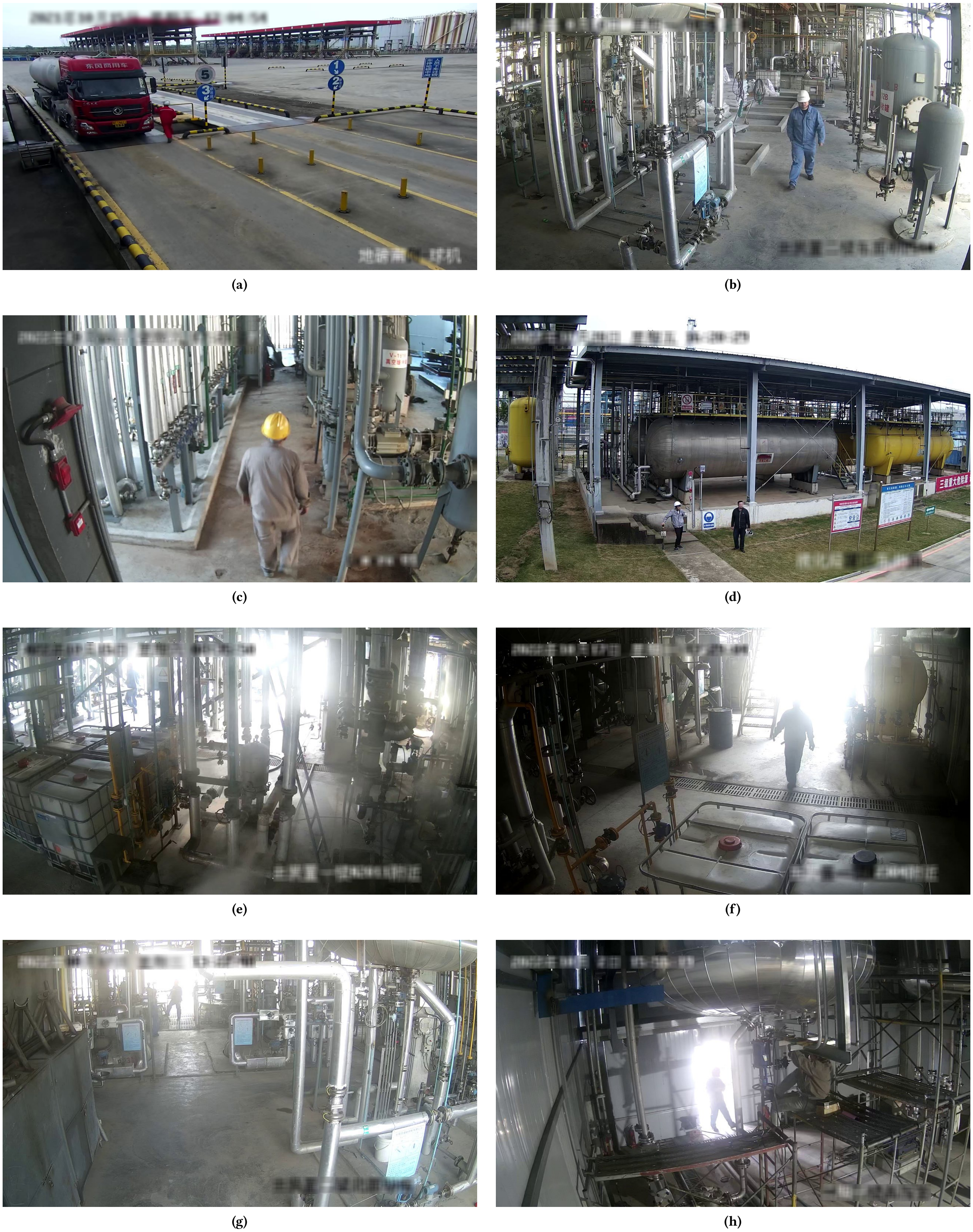
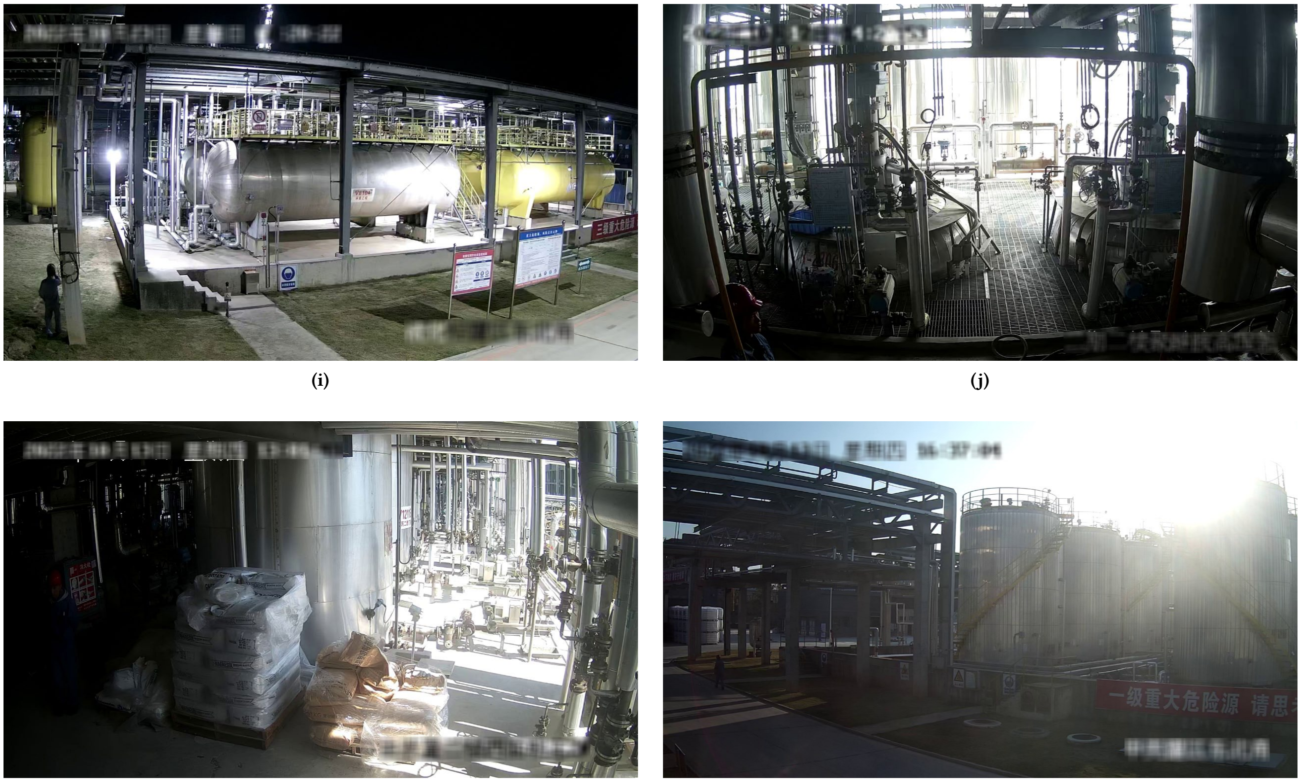


Figure 2: Image examples with different lighting conditions.



(i) (j) (k) (l)

Figure 2: Image examples with different lighting conditions.

one of the most challenging object detection datasets available, including two commonly used versions, COCO 2005 and COCO 2018. MS-COCO contains 80 categories, whose annotated objects are divided into three categories based on scale: small objects, medium objects, and large objects. The ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) is also a competition that includes an object detection task, and its dataset serves as a benchmark for the evaluation of algorithm performance. The number of objects in the LISVRC was expanded to 200 compared to the VOC dataset, and the number of images and annotated objects are significantly larger than those in the VOC dataset. The Open Image dataset was launched by Google in 2017 and consists of 9.2 million images with 16 million bounding boxes for 600 categories on 1.9 million images used for detection, becoming one of the largest object detection datasets.

Despite the multitude of datasets [6, 7, 17, 20, 34] available for the field of object detection, there is a relative dearth of datasets specifically tailored to industrial scenarios. The availability of high-quality and authentic datasets for safety clothing and helmet detection is crucial for ensuring engineering safety and factory compliance. Although object detection algorithms have made significant strides, their widespread application in industrial settings is still limited due to the absence of high-quality datasets. To our knowledge, there are currently only two open-source datasets available for helmet detection, namely Pictor-v3 [25] and SHWD [12]. However, the utility of these datasets is constrained, and they struggle to meet the application demands of both academia and industry. The Pictor-v3 dataset contains only 1,330 annotated images, which is insufficient to meet the contemporary requirement for large-scale data in deep learning. Moreover, the dataset is deficient in the inclusion of negative samples, leading to an imbalanced data composition and a disproportionate emphasis on color labels. In reality, safety helmets come in a variety of colors, and an overemphasis on color labels could potentially impair the model's ability to recognize the physical attributes of the helmets. The SHWD dataset comprises 3,241 images related to safety helmets, with 10,457 instances annotated. Although there is a marked increase in the number of images compared to Pictor-v3, most images in SHWD are downloaded from the web, featuring relatively simple backgrounds that do not adequately represent the complexity of real-world scenarios. Furthermore, due to the diverse origins of the data, there is a significant variation in image resolution within the SHWD dataset, which may adversely affect the model's performance in practical applications. In this work, we collect a high-quality and large-scale dataset, SFCHD, that authentically reflects the complexity of industrial scenarios, to facilitate the application of object detection algorithms in the field of industrial safety.

B.2 Object Detection Method

In the field of computer vision, object detection is a fundamental task for many real-world applications, which are receiving widespread attention. There are many potential application scenarios for object detection, such as autonomous driving [9, 40], medical diagnosis [5], emotion detection [18], pedestrian detection [15, 33], etc. The early object detection algorithms are composed of handcrafted feature extractors such as Histogram of Oriented Gradients (HOG) [4], Viola Jones detector [38], and so on. These methods have inaccurate detection results and are resistant to be applied to different datasets. In recent years, as the rapid evolution of deep learning, the reintroduction of CNN based models has brought a new landscape to the realm of object detection. CNN-based [16, 41] object detection methods can be categorized into two main types: two-stage methods and one-stage methods. Two-stage algorithms [10, 11, 27, 32] have two separate steps, i.e., searching for object proposals in images during the first stage, and then classifying and localizing them during the second stage. These methods commonly are of complex architecture and demand longer time to generate proposals. Unlike the two-stage algorithms, one-stage approaches [1, 21, 29–31, 35] are superior in terms of real-time performance and network architecture. One-stage algorithms localize objects with the predefined boxes of different scales and aspect ratios, and classify semantic objects in a single shot adopting dense sampling [43].

B.3 Low-Light Object Detection

In low-light environments, the visibility of the image is severely compromised due to poor lighting conditions, greatly impacting the performance of detector. Initial methods for low-light detection focus on enhancing the low-light images before feeding them into detector. For instance, Zhang et al. [45] introduced KinD, which achieves image enhancement through paired training with images at different levels of illumination. Lv et al. [24] proposed a multi-branch low-light enhancement network (MBLLEN), which extracts features at various levels and then merges them using a multi-branch structure to produce enhanced images. Liu et al. [22] proposed IA-YOLO, which uses a CNN-based parameter predictor to train the YOLO detector. Qin et al. [28] introduced DE-YOLO, by training the cascaded DENet and YOLOv3 models in an end-to-end manner with the standard detection loss. Cui et al. [2] presented IAT-YOLO, which utilizes both local and global branches, incorporating transformers for image enhancement. Additionally, joint training methods have been employed, where only the loss from detection tasks is used to accomplish low-light detection.

C PROPOSED SCALE MODULE

To address the challenges faced by object detection models in low-light environments, we design a spatial and channel attention-based low-light enhancement (SCALE) module. SCALE, as a plug-and-play component, integrates seamlessly into traditional object detection workflows, as depicted in Fig. 3. Specifically engineered for extreme low-light conditions, the SCALE module adaptively enhances images, consisting primarily of two components, i.e., spatial attention pathway (SAP) and channel attention pathway (CAP). SAP enhances the network's ability to focus on important areas of the image by assigning differentiated weights to pixels in various locations, thereby more effectively understanding and capturing the spatial structure and its correlations within the image. The CAP, on the other hand, guides the network to concentrate on the most informative and significant channels by assigning weights, while suppressing features from redundant channels, further improving the quality and accuracy of feature expression. Features processed by the SCALE module are then fed into the object detector. The entire pipeline relies solely on detection loss for end-to-end training, simplifying the training process and enhancing efficiency. This design not only improves the model's performance under low-light conditions but also maintains the simplicity of model training.

C.1 Spatial Attention Pathway

SAP enhances the network's ability to focus on important areas of the image by assigning differentiated weights to pixels in various locations, thereby more effectively understanding and capturing the spatial structure and its correlations within the image. Firstly, the low-light image I is initially input into three convolutional kernels of varying sizes to extract features from multiple scales. Subsequently, the obtained feature maps are concatenated along the channel dimension to enrich the feature space. This process can be mathematically represented as:

$$F_k = \mathrm{Conv}_k(I), F_{Conv} = \mathrm{Cat}[F_1, F_3, F_5], \tag{1}$$

where $k \in \{1, 3, 5\}$, and I denotes an input with a resolution of $H \times W \times C$; Conv_k denotes the convolutional operation with a kernel of size k, e.g., Conv_1 is the convolution with a kernel of size 1; F_k denotes the corresponding convolution branch output; Cat is the concatenation operation across the channel dimension; F_{Conv} denotes the feature map after passing through parallel convolution branches.

The obtained feature map F_{Conv} is then directed into two independent spatial attention units (i.e., SA in Fig. 3). Each Spatial Attention unit primarily consists of pooling, convolution, and element-wise multiplication operations. Initially, we apply global max pooling and global average pooling to the feature map. The global max pooling helps the network to filter the most salient feature, focusing on areas of the low-light image that contain key information while disregarding the effects of darkness and noise. The global average pooling smooths the feature value, making the extracted feature more stable and representative. Subsequently, these two pooled feature maps are concatenated, fed into a convolution for fusion, and normalized through a sigmoid function. Ultimately, the normalized fused feature is element-wise multiplied with each channel of the original feature map, yielding the weighted spatial attention feature maps F_{SA_k} ($k \in \{3, 5\}$). The process above can be mathematically expressed as:

$$F_{SA_k} = F_{Conv} \otimes \sigma(\mathrm{Conv}_k(\mathrm{Cat}[F_{Max}, F_{Avg}])), \tag{2}$$

where F_{Max} and F_{Avg} represent the values after applying max pooling and average pooling to F_{Conv}, respectively; σ represents the sigmoid activation function, and $\otimes$ is the element-wise multiplication. The spatial attention feature maps F_{SA_3} and F_{SA_5} are concatenated, then passed through a convolution of size 3 to produce the output

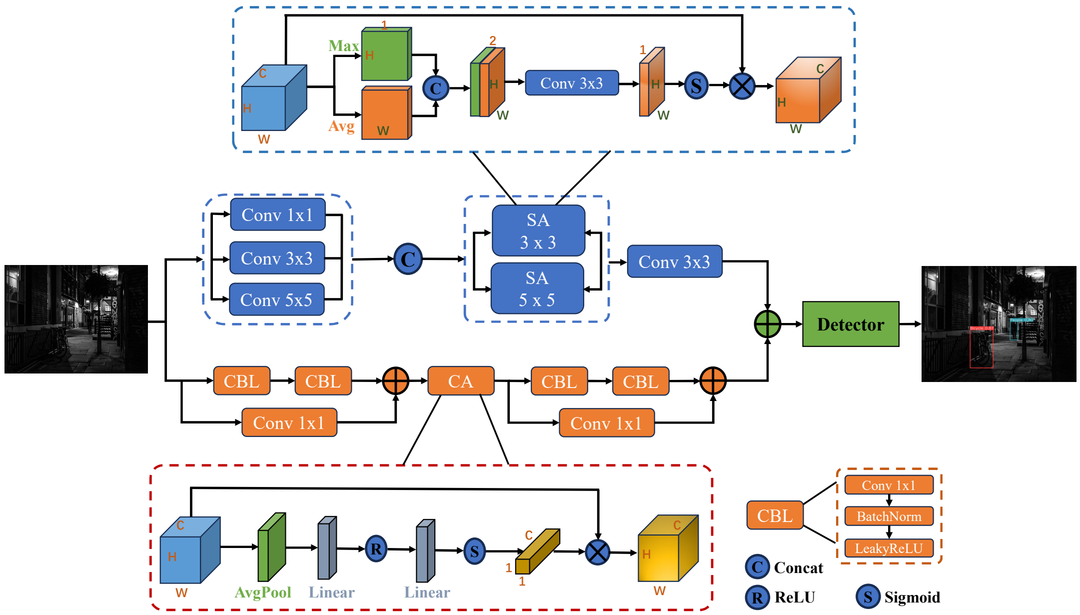


Figure 3: Network structure of spatial and channel attention-based low-light enhancement (SCALE) module.

of SAP:

$$\begin{aligned} F'_{Conv} &= \mathrm{Conv}_3(\mathrm{Cat}[F_{SA_3}, F_{SA_5}]), \\ I_{SAP} &= I \otimes F'_{Conv}. \end{aligned} \tag{3}$$

C.2 Channel Attention Pathway

The CAP guides the network to concentrate on the most informative and significant channels by assigning weights, while suppressing features from redundant channels, further improving the quality and accuracy of feature expression. Before entering the channel attention unit (CA in Fig. 3), the low-light image I is first input into a residual block. This residual block primarily consists of convolution, batch normalization, and residual operations, which can be mathematically expressed as:

$$\begin{aligned} \hat{F}_{Conv} &= \mathrm{Conv}_1(I), F_{CBL} = \mathrm{LR}(\mathrm{BN}(\mathrm{Conv}_1(I))), \\ F_{Res} &= \hat{F}_{Conv} + \mathrm{LR}(\mathrm{BN}(\mathrm{Conv}_3(F_{CBL}))), \end{aligned} \tag{4}$$

where Conv_1 and Conv_3 denote convolution operations with kernel sizes of 1 and 3, respectively; LR represents the LeakyReLU activation function, and BN denotes batch normalization; a residual block comprises two CBL blocks, and each CBL block includes a convolution layer, LeakyReLU function, and batch normalization; F_{CBL_1} signifies the feature map after the first CBL block, and F_{Res} indicates the final output of the residual block.

Inspired by SENet [14], the obtained feature map F_{Res} is then directed into the channel attention unit. This unit primarily consists of pooling, linear transformation, and element-wise multiplication operations. Initially, an average pooling operation is employed to down-sample the feature map. This operation not only reduces the complexity of the feature map but also diminishes noise and detail interference, capturing local statistical information. Subsequently, the pooled feature map undergoes two linear transformations and is normalized through a sigmoid function. Ultimately, the resulting feature map is element-wise multiplied with the original feature map across channels to yield the channel attention feature map F_{CA}. This process assigns different weights to each feature channel, enabling the network to learn the most informative feature channels. The computation can be mathematically represented as:

$$F_{CA} = F_{Res} \otimes \sigma(\mathrm{Lin}_2(\mathrm{ReLU}(\mathrm{Lin}_1(F'_{Avg})))), \tag{5}$$

where F'_{Avg} denotes the value after applying the average pooling operation to F_{Res}; Lin_1 and Lin_2 represent two successive linear transformations. F_{CA} is processed through a residual block to obtain the output of CAP:

$$\begin{aligned} \tilde{F}_{Conv} &= \mathrm{Conv}_1(F_{CA}), \tilde{F}_{CBL} = \mathrm{LR}(\mathrm{BN}(\mathrm{Conv}_1(F_{CA}))), \\ \tilde{F}_{Res} &= I \otimes \tilde{F}_{Conv} + \mathrm{LR}(\mathrm{BN}(\mathrm{Conv}_3(\tilde{F}_{CBL}))), \\ I_{CAP} &= I \otimes \tilde{F}_{Res}. \end{aligned} \tag{6}$$

D EXPERIMENTAL SETUP

Centered around the SFCHD dataset and the SCALE module, our experiments are composed of three parts. Initially, we utilize a variety of classic detection methods to perform experiments on the existing helmet datasets and our SFCHD dataset, validating the practicality of our dataset. Subsequently, we incorporate the SCALE module into detection algorithms and conduct experiments on the public low-light dataset to confirm the effectiveness of the SCALE module. Finally, we integrate the SCALE module with the SFCHD dataset, demonstrating its capability to significantly improve detection accuracy under challenging conditions.

D.1 Usability of SFCHD Dataset

In this subsection, we employ several detection models to conduct experiments on the existing helmet datasets and proposed SFCHD dataset, reporting their outcomes to substantiate the usability of our dataset.

Datasets: We compare the propose SFCHD with the Pictor-v3 [25] and SHWD [12] datasets. Pictor-v3 contains 774 crowd-sourced and 698 web-mined images, where the crowd-sourced and web-mined images contain 2,496 and 2,230 instances of workers, respectively. SHWD provides the dataset used for both safety helmet wearing and human head detection. It includes 7,581 images with 9,044 human safety helmets wearing objects (positive) and 111,514 normal head objects (not wearing or negative).

Methods: We utilize classic detection methods to verify the usability of our dataset, including SSD [21], Faster RCNN [32], RetinaNet [19], FCOS [35], TOOD [8], VFNet [44], YOLOv5 [36]. SSD was a one-stage, multi-scale detection algorithm that utilized a convolutional pyramid for feature extraction and performed classification and regression across feature maps. Faster RCNN was a pioneering two-stage object detection framework that identified potential object regions, or anchor boxes, and then precisely classified and localized the objects within these regions. RetinaNet introduced the Focal Loss to address the class imbalance in one-stage detectors, enhancing performance without altering the underlying network structure. FCOS redefined object detection by offering an anchor-free and proposal-free approach, operating pixel-by-pixel for a streamlined, full convolutional network-based solution. TOOD proposed a task-aligned one-stage detection method with a novel head structure and learning method, focusing on improving detection for specific tasks. VFNet designed the Varifocal Loss function to predict IoU-aware classification scores and introduced a star-shaped bounding box feature representation for refining object localization. YOLOv5 was a one-stage object detection algorithm, improved upon its predecessors by offering superior speed and accuracy in one-stage object detection.

Details: All experiments are conducted using MMDetection [26], with dataset partitions following the default configuration. The experiments are optimized using the SGD optimizer. The momentum and weight decay are set to 0.9 and 0.0001, respectively. During the training and testing process, the image size was adjusted to 1333 * 800. The batch size is set to 2 for a single GPU, and distributed training is conducted using 4 GPUs. All models are trained on NVIDIA GeForce RTX 3090 GPUs, each with 24GB of graphics memory. We evaluate the mean average precision (mAP) of the object detection to measure the performance of all models fairly. The mAP(0.50:0.95) metric measures the mAP across different Intersection over Union (IoU) thresholds, ranging from 0.50 to 0.95 with an interval of 0.05. Similarly, mAP(0.50) is the mAP at a single IoU threshold of 0.50.

D.2 Validity of SCALE Module

In this subsection, we integrate the SCALE module with detection algorithms and experiment on the public low-light dataset to validate the efficacy of SCALE.

Dataset: To substantiate the efficacy of the SCALE module, we conduct extensive experiments on the ExDark [23] dataset. The dataset is a benchmark specifically designed to assess object detection algorithms under low-light conditions. It comprises a total of 7,363 low-light images, categorized into 10 varying illumination conditions, ranging from extremely dim to moderately low light, across 12 classes, with a total of 23,710 annotated instances. The dataset is partitioned into a training set with 5,896 images and a testing set with 1,467 images. The images within the dataset encompass a multitude of scenarios from indoor to outdoor settings and from urban to natural landscapes, ensuring the diversity and applicability of the dataset.

Methods: The models we select primarily encompass three categories. The first category consists of low-light image enhancement methods, including KinD [45], MBLLEN [24], and Zero-DCE [13]. KinD, inspired by Retinex theory, decomposed the image into two components, one responsible for light regulation (illumination) and the other for degradation elimination (reflectance). MBLLEN extracted rich features at different levels, enhanced them through multiple sub-networks, and ultimately produced the output image through multi-branch fusion. Zero-DCE formulated the task of light enhancement as a specific curve estimation for images with a deep network, training a lightweight deep network to estimate the dynamic range adjustment of pixels and higher-order curves for a given image. The second category is low-light image object detection methods, including MAET [3], DENet [28], IAT-YOLO [2], and PE-YOLO [42]. MAET, in a self-supervised manner, considered the physical noise model and image signal processing to learn the intrinsic visual structure by encoding and decoding real-world illumination degradation transformations. DENet enhanced the model's performance under adverse weather conditions by using a Laplacian pyramid to decompose each input image into a low-frequency component and several high-frequency components. IAT-YOLO proposed a lightweight and fast illumination adaptive transformer for normalizing sRGB images under low-light or underexposed/overexposed conditions. PE-YOLO introduced a detail processing module composed of a context branch and an edge branch to enhance image details, as well as a low-frequency enhancement filter to obtain low-frequency semantic information and avoid high-frequency noise. The third category consists of classical one-stage object detection methods, including FCOS [35], VFNet [44], and TOOD [8]. These methods are primarily utilized for cascading with the SCALE module to form new end-to-end models.

Details: The inaugural category of models is trained on the LOL [39] dataset, subsequently employing the trained models to process the ExDark dataset, thereby generating enhanced and brightened imagery. These images are then subsequently input into the YOLOv3 [31] detector to procure the detection outcomes of the low-light image enhancement techniques. In contrast, the second category of models directly integrates low-light images into the network for training, culminating in the ultimate detection results. To ensure a fair comparative analysis, the detector employed across both categories of methods is the YOLOv3. Throughout the training and testing phases, the dimensions of the images are uniformly adjusted to 608 * 608. For training, the batch size for an individual GPU is configured to 4, and distributed training is executed across 4 GPUs. The optimization algorithm utilized during training is SGD, with the learning rate set to 0.001 and the weight decay factor designated as 0.0005. The third category of methods is to validate the enhancement effect of the SCALE module under low-light conditions. We integrate the SCALE module with existing object detection algorithms to obtain the detector's output results. In this category, experiments are conducted based on the MMDetection framework. During training and

Method	Bicycle	Boat	Bottle	Bus	Car	Cat	Chair	Cup	Dog	Motorbike	People	Table	mAP(0.50)
	AP(0.50)												
YOLOv3 [31]	79.8	75.3	78.1	92.3	83.0	68.0	69.0	79.0	78.0	77.3	81.5	55.5	76.4
KinD [45]	80.1	77.7	77.2	93.8	83.9	66.9	68.7	77.4	79.3	75.3	80.9	53.8	76.3
MBLLEN [24]	82.0	77.3	76 .5	91.3	84.0	67.6	69.1	77.6	80.4	75.6	81.9	**58.6**	76.8
Zero-DCE [13]	84.1	77.6	78.3	93.1	83.7	70.3	69.8	77.6	77.4	76.3	81.0	53.6	76.9
MAET [3]	83.1	78.5	75.6	92.9	83.1	73.4	71.3	79.0	79.8	77.2	81.1	57.0	77.7
DENet [28]	80.4	**79.7**	77.9	91.2	82.7	72.8	69.9	80.1	77.2	76.7	82.0	57.2	77.3
IAT-YOLO [2]	79.8	76.9	78.6	92.5	83.8	73.6	72.4	78.6	79.0	**79.0**	81.1	57.7	77.8
PE-YOLO [42]	**84.7**	79.2	**79.3**	92.5	83.9	71.5	71.7	79.7	79.7	77.3	81.8	55.3	78.0
SCALE-YOLO (Ours)	81.3	79.3	78.2	**93.9**	**84.2**	**75.5**	**74.9**	**82.3**	**81.0**	77.5	**82.5**	57.3	**79.0**

Table 1: Performance comparisons between our SCALE-YOLO and existing models on the ExDark dataset. These models include both low-light image enhancement methods and low-light image object detection methods.

Method	Bicycle	Boat	Bottle	Bus	Car	Cat	Chair	Cup	Dog	Motorbike	People	Table	mAP(0.50)
	AP(0.50)												
FCOS [35]	75.5	64.4	68.0	86.8	78.5	69.3	55.4	71.7	70.0	64.8	72.3	46.7	68.6
FCOS+SCALE	75.1	66.6	73.5	89.9	78.9	67.0	57.2	72.8	74.2	67.3	72.0	45.9	**70.0**
VFNet [44]	77.4	70.5	76.6	90.6	81.8	67.1	59.4	71.8	72.6	70.6	77.7	53.3	72.5
VFNet+SCALE	79.4	70.2	76.5	89.7	81.7	71.9	60.9	71.5	75.0	71.2	77.4	55.5	**73.4**
TOOD [8]	77.0	69.2	72.2	90.0	80.0	72.6	63.0	71.8	71.0	71.9	76.2	52.2	72.3
TOOD+SCALE	77.9	70.0	78.3	90.0	80.7	69.1	62.0	72.4	73.7	69.2	78.1	54.2	**73.0**

Table 2: Performance improvements of the SCALE module on the ExDark dataset.

Method	SAP	CAP	mAP(0.50)
YOLOv3 [31]	–	–	76.4
SCALE-YOLO (Ours)	✓	✗	77.3
	✗	✓	77.8
	✓	✓	**79.0**

Table 3: Ablation analysis for different pathways in SCALE.

testing, the size of the images is uniformly adjusted to 1333 * 800. The batch size for a single GPU is set to 4, and distributed training is performed using the same 4 GPUs. The optimizer for training is the SGD, with the learning rate set to 0.01 and the weight decay factor set to 0.0001.

Results: To substantiate the efficacy of the SCALE module, we conduct extensive experiments on the ExDark [23] dataset, a publicly available low-light benchmark. Table 1 compares the performance of our proposed SCALE-YOLO with existing models encompassing both low-light image enhancement methods and low-light image object detection methods. SCALE-YOLO outperforms YOLOv3 by 2.6% in mAP. It also surpasses image enhancement models KinD, MBLLEN, and Zero-DCE, which though enhance brightness and details, can amplify noise and degrade feature extraction, by 2.7%, 2.2%, and 2.1% respectively. The table indicates that object detection methods generally outperform image enhancement methods, suggesting that understanding image context is more effective than just increasing contrast and brightness. SCALE-YOLO improves mAP by 1.3% over MAET, 1.7% over DENet, 1.2% over IAT-YOLO, and 1% over PE-YOLO. SCALE-YOLO leverages spatial attention to capture regional correlations and channel attention to focus on significant feature channels in low-light images, thereby enhancing detection performance.

Table 2 displays the performance improvements when the SCALE module is integrated into one-stage object detection algorithms on the ExDark dataset. The integration leads to notable enhancements: FCOS+SCALE achieves a 1.4% mAP increase over FCOS, VFNet+SCALE realizes a 0.9% mAP boost, and TOOD+SCALE improves by 0.7% over TOOD. These results confirm that the SCALE module strengthens detectors' comprehension and recognition under low-light conditions, subsequently elevating their detection accuracy.

To assess the contributions of the spatial and channel attention pathways within the SCALE module, we perform ablation studies as detailed in Table 3. The results show that by activating only the SAP, the model improves the mAP by 0.9% compared to YOLOv3, validating its ability to concentrate on salient information in low-light images. Enabling only the CAP increases the mAP by 1.8%, indicating its role in highlighting informative channels. Comparatively, inter-channel relationships seem to be more influential than spatial cues. When both SAP and CAP are combined, the model's mAP improves by 2.6%. These findings underscore the SCALE module's ability to generate more discriminative features, enhancing detectors' comprehension of low-light imagery.

D.3 Application of SCALE on SFCHD

In this subsection, we apply the SCALE module to the SFCHD dataset to demonstrate that SCALE can effectively enhance the detection quality on the SFCHD dataset.

To verify the performance of the SCALE module on the SFCHD dataset, we select a range of object detection models for testing, including FCOS [35], VFNet [44], TOOD [8], and YOLOv8 [37]. We integrate the proposed SCALE module into these detectors to construct an end-to-end low-light object detection model, which is optimized solely by a single loss function. In terms of experimental setup, we continue with the configuration from previous experiments, utilizing distributed training across 4 GPUs and employing the SGD optimizer. The learning rate is uniformly set to 0.01. For the FCOS, VFNet, and TOOD models, the input image size is adjusted to 1333 * 800 pixels, the Batch Size per GPU is 2, and the weight decay factor is 0.0001. For YOLOv8, the image size is adjusted to 640 * 640 pixels, the Batch Size per GPU is increased to 16, and the weight decay factor is set to 0.0005.

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